Predictive UFLS Ride-Through Extension via PMU Wavefront Analysis

A Gaussian Process Learning Framework for Inertia-Conditioned Protection Coordination

A Position Paper

Version 0.8 — DRAFT

October 2026

Timothy E. Roxey

Eclectic Technologies

T. Roxey, "Predictive UFLS Ride-Through Extension via PMU Wavefront Analysis: A Gaussian Process Learning Framework for Inertia-Conditioned Protection Coordination," Version 0.8, Eclectic Technologies, October 2026.

Concept DOI: https://doi.org/10.5281/zenodo.22818437

Version DOI: https://doi.org/10.5281/zenodo.23439571

Current version:
https://scubanuke.github.io/publications/papers/Predictive_UFLS_Ride_Through.pdf

Draft. The conjecture is unvalidated; Section 6 proposes how it could be tested.

Abstract

Conventional frequency ride-through protection for generating units is calibrated to fixed thresholds under NERC Reliability Standard PRC-024, without access to real-time information about how long a frequency excursion will last at a given location. When the excursion would resolve within the ride-through window, units may trip unnecessarily, deepening the disturbance rather than riding through it. This paper advances two contributions. First, it establishes that the Full Width at Half Maximum (FWHM) of the frequency wavefront, as observed at a local phasor measurement unit (PMU), encodes the predicted duration of the frequency excursion at that location — directly, in the same units, from the same PMU time-series. Second, it proposes a Gaussian process regression architecture that learns the local FWHM-to-duration mapping from accumulated event data, produces explicit uncertainty estimates, and activates ride-through extension only when the predicted excursion duration falls within the PRC-024 ride-through window under worst-case posterior uncertainty. PMU-to-PMU observation sharing via the existing IEEE C37.118 layer expands the calibration dataset without introducing real-time latency into the protection decision. The 2008 FPL Flagami event motivates the work, and a validation method is proposed for testing the conjecture against it; no simulation has yet been run. A pathway for revision of NERC PRC-024 to accommodate model-based alternative compliance is proposed.

Erratum to Version 0.7

Version 0.7 of this paper, deposited under version DOI 10.5281/zenodo.22818438 (concept DOI 10.5281/zenodo.22818437), said in Section 4.6 that the ride-through extension satisfies all three conditions of local sovereignty in the Foundational Definition FD-LS: “The architecture meets all three.” That statement is withdrawn. The extension decides on a learned Gaussian process posterior, and FD-LS, as issued in the October 2026 Edition, requires that a locally sovereign function’s decision contain no learned or stochastic element. Section 4.6 now separates the local execution the architecture does achieve from local sovereignty, which the fixed-threshold floor beneath the extension can carry and the extension cannot, and Section 7.3 and the Conclusion follow it. The physical argument, the learning architecture and the standards pathway are otherwise unchanged. Version 0.7 stands as deposited, a record of what was said on its date.

1. Introduction

The Flagami substation in Miami is not unfamiliar to this paper’s author. In the early 1980s, he worked at Florida Power & Light’s Miami General office — close enough to Flagami to know the substation by name, though not by any direct engineering involvement. He moved on to AEP and then to BGE, and was at BGE when the February 2008 disturbance occurred. He came to

NERC in 2009. It was there, around 2009 or 2010, while he and a NERC colleague were working on the Aurora alerts, that the conversation took place. They were looking at the South Florida frequency records — the PMU data showing the disturbance propagating outward across the Eastern Interconnection, attenuating as it traveled through TVA and into Ontario, and reaching western Massachusetts [11]. The image that fixed itself was a bell: strike it at one point, and the vibration travels through the material, broadening and attenuating as it goes. The grid rang the same way.

The question they posed: if you could observe the shape of that ring at a local generating unit — whether the frequency excursion arriving at that location was sharp and brief or broad and already attenuating — could you tell the protection relay whether it actually needed to trip? If the excursion was broad at the local PMU, the unit might be sitting two seconds from riding through when the protection relay fired unnecessarily. That question has percolated for fifteen years. The analytical tools to answer it rigorously are now available.

The protection coordination problem at the heart of this paper is not exotic. It follows directly from what frequency protection relays do and do not know at the moment they make a trip decision.

A generating unit connected to the transmission system carries frequency protection settings consistent with NERC Reliability Standard PRC-024. Those settings define a frequency-versus-time envelope: if the local frequency falls below a specified threshold and remains there for a specified duration, the unit trips. The settings are fixed. They are conservatively calibrated — for the worst-case conditions the grid might present — because the relay has no way to know how long the local excursion will last when it makes the trip decision.

That conservatism has a cost. When the frequency excursion would resolve within the ride-through window — when the grid, left alone, would absorb the disturbance before the unit's PRC-024 time limit expires — units may trip anyway. Their trips further reduce generation, deepen the frequency excursion, trigger additional trips, and potentially convert a recoverable disturbance into a cascade. The relay's decision, made in ignorance of the excursion's duration, can make the grid less stable than it would have been had the relay known what the PMU was already recording.

This is the protection coordination gap. It is not a gap in relay hardware capability — modern digital relays can execute sophisticated algorithms in milliseconds. It is an information gap. The relay does not know how long the local frequency excursion will last when it needs to make the trip decision.

Two developments have changed the analytical landscape sufficiently to warrant revisiting this gap. The first is the widespread deployment of phasor measurement units across the transmission system. PMUs sample frequency, voltage, and current at rates of 30 to 60 frames per second, time-stamped to UTC via GPS. The temporal resolution is sufficient to observe the shape of a frequency wavefront as it arrives at a measurement point — not just the frequency value, but the

wavefront's time-domain structure, including its width. The second is the maturation of machine learning methods for time-series regression that can operate on PMU data streams with explicit uncertainty quantification. The combination of high-resolution local measurement and a learning architecture that improves with experience opens a pathway to information that the relay previously lacked.

This paper proposes that the Full Width at Half Maximum (FWHM) of the frequency wavefront — a shape characteristic computable from the existing PMU time-series without additional instrumentation — predicts the duration of the local frequency excursion below the PRC-024 ride-through threshold. FWHM and excursion duration are both measured in seconds from the same PMU recording; the GP model learns the local mapping between them from accumulated event data, without requiring a closed-form derivation for the specific network topology.

The Flagami event in February 2008 serves as the motivating case study. A 345 kV bus fault at the Flagami substation in Miami initiated a cascade of generating unit trips, including Turkey Point nuclear units 3 and 4, through a sequence of frequency protection operations. R. W. Cummings documented the wave-like propagation of the resulting frequency disturbance across the Eastern Interconnection using PMU data — the first such observation on an interconnection-wide basis [11]. The paper's working premise, still to be confirmed against the post-event record, is that some of the tripped units could have ridden through had protection settings permitted it. Section 6 proposes how a reconstruction of the Florida network could test whether the FWHM of the wavefront at the tripped-unit locations is consistent with the predicted excursion duration.

A third development reinforces the paper's timeliness. The rapid penetration of inverter-based resources — utility-scale wind, solar photovoltaic, and battery storage operating with grid-following inverter controls — is reducing and destabilizing aggregate synchronous inertia across all North American interconnections. Simultaneously, the growth of hyperscale computational loads — data center campuses drawing hundreds of megawatts from the transmission system — is introducing a new class of large, abrupt imbalance events, in both directions, that the frequency protection framework was not designed to accommodate. NERC's Level 3 Essential Action Alert of 4 May 2026 on computational loads, and FERC's order of 16 July 2026 directing NERC to file reliability standards and registration criteria for computational load integration, have made that load class a formal reliability concern. The consequence is an environment where effective inertia is lower, more variable by hour and season, and more consequential for individual frequency events than it was when PRC-024 was calibrated. The case for duration-conditioned ride-through extension strengthens as that mismatch grows.

The standards implication is a proposed revision pathway for NERC PRC-024 that accommodates a model-based alternative compliance path alongside the existing fixed-envelope approach. The near-term deliverable is a NERC Standards Authorization Request (SAR). Section 7 develops the standards pathway and identifies the appropriate standards development vehicles.

2. Background

2.1 Frequency Wavefront Propagation

A generation loss event does not produce an instantaneous, spatially uniform frequency response across an interconnected grid. The frequency perturbation propagates as an electromechanical wave, traveling outward from the disturbance origin through the network of synchronous machines and transmission links. The propagation speed and attenuation characteristics depend on the distribution of rotating machine inertia, network impedance, and load characteristics throughout the grid. At locations near the disturbance's origin, the wavefront arrives quickly, and the frequency deviation is sharp. At more distant locations, the wavefront arrives later and, if the intervening inertia is sufficient, with a reduced rate of change and a broader time-domain shape.

This spatial and temporal structure of the frequency response has been documented in the PMU literature since the early deployment of wide-area measurement systems. The Western Interconnection, with its large geographic extent and heterogeneous inertia distribution, provides the most extensively studied examples. Electromechanical wave propagation speeds on the order of hundreds to thousands of kilometers per second have been estimated from PMU event recordings, with attenuation characteristics that reflect the aggregate inertia along the propagation corridor.

2.2 Rate of Change of Frequency as an Inertia Estimator

The conventional approach to inertia estimation in protection and monitoring applications uses ROCOF — the Rate of Change of Frequency, df/dt — measured at the local PMU or derived from the relay’s frequency tracking. The relationship between ROCOF and inertia follows from the swing equation. For a machine observing a generation-load imbalance ΔP, the initial ROCOF is inversely proportional to the effective system inertia constant H. This relationship is analytically grounded and has been the basis for inertia estimation in grid studies.

In practice, ROCOF estimation in the first seconds of a disturbance is challenged by three factors. First, electromechanical oscillations superimposed on the frequency trajectory introduce measurement noise, rendering instantaneous ROCOF estimates unreliable without filtering. Second, the filtering required to stabilize ROCOF estimates introduces delay — typically 100 to 500 milliseconds — during which the protection decision may need to be made. Third, ROCOF is a scalar derived from the local frequency time series and contains no spatial information about how the disturbance is propagating through the network. As a result, ROCOF-based inertia estimation is least reliable precisely when it is most needed: in the first fraction of a second of a severe disturbance.

2.3 FWHM in Signal Processing Context

Full Width at Half Maximum is a standard measure of pulse width in signal processing, spectroscopy, and optics. For a pulse or peak, FWHM is the duration of the pulse at half its

maximum amplitude. It is a shape characteristic, not an amplitude characteristic — two pulses with identical maximum amplitude but different widths have different FWHM values.

Applied to a frequency wavefront, FWHM is the duration of the frequency deviation event — measured from the local PMU frequency time-series — at half the maximum frequency deviation from nominal. Two disturbances with the same maximum frequency deviation but different propagation histories will produce different FWHM values at a given observation point. A disturbance traveling through a high-inertia corridor produces a broader, slower wavefront — larger FWHM — than the same disturbance traveling through a low-inertia corridor. FWHM is directly computable from the existing PMU frequency time-series, requires no additional instrumentation, and does not require knowledge of the disturbance origin location or the network topology between origin and observer.

2.4 UFLS and the Cascade Dynamic

Under-Frequency Load Shedding (UFLS) is the last-resort defense against frequency collapse. When frequency falls below programmed thresholds, UFLS relays automatically disconnect blocks of load, reducing demand to arrest the decline in frequency. UFLS is designed as a backstop — it operates after the primary frequency response from spinning reserve and governor action has had time to take effect. It sheds load in blocks, in sequence, each block triggered at a successively lower frequency threshold.

The interaction between the generating unit frequency protection trips and the UFLS defines the cascade dynamics. When generating units trip on frequency protection, they reduce supply further — deepening the frequency excursion and activating additional UFLS blocks. If generating unit trips are unnecessary — if the unit would have remained within its ride-through envelope had inertia conditions been correctly assessed — those trips consume UFLS headroom and widen the outage without providing protective benefit. Preventing unnecessary protection trips in high-inertia conditions directly reduces UFLS demand and improves the probability of frequency recovery.

2.5 Inverter-Based Resources, Large Computational Loads, and the Inertia Variability Problem

The frequency protection coordination problem described in this paper is not static. Two structural trends in the bulk electric system are simultaneously making it more severe and more consequential.

The first is the large-scale displacement of synchronous generation by inverter-based resources (IBRs). Utility-scale wind and solar photovoltaic plants, and battery storage systems operating under grid-following inverter controls, do not contribute synchronous inertia to the interconnection. A grid-following IBR observes the grid frequency and modulates its power output accordingly; it does not exchange kinetic energy with the grid through the

electromagnetic coupling of a rotating mass. As IBR penetration increases and synchronous generators are decommissioned or operated less frequently, aggregate system inertia decreases and — crucially — becomes more variable. Inertia is now a function of dispatch conditions: high during periods of synchronous-machine-dominated dispatch, low during periods of high renewable penetration. A given generating unit at a given network location may face dramatically different effective inertia conditions at different hours of the day or different seasons of the year. Fixed PRC-024 settings, calibrated to a single conservative envelope, cannot track this variability.

Grid-forming inverter technology offers a partial response to this problem. A grid-forming IBR synthesizes an inertia-like response — it emulates the kinetic energy exchange of a synchronous machine through fast power electronics control — and can contribute to frequency support in ways that grid-following IBRs cannot. Grid-forming deployments are growing but remain a small fraction of total IBR capacity. Their contribution to inertia is synthetic rather than mechanical, and its interaction with the electromechanical wave propagation that produces the FWHM signal is a refinement addressed in Section 3.2.

The second trend is the emergence of hyperscale computational loads as a transmission-connected load class. Data center campuses drawing 100 to 500 megawatts from the bulk electric system are now interconnecting at transmission voltage in multiple North American markets, including ERCOT. These facilities can connect, disconnect, or shift load in large blocks over timescales of seconds to minutes — timescales directly relevant to frequency stability. A facility that drops off the grid is, from the grid's perspective, the mirror image of a generating unit trip: losing load drives frequency up rather than down. A block of load that returns uncoordinated, or a large workload that migrates onto a campus, drives it down. Either way the frequency protection system sees a disturbance and responds accordingly.

Two events have shown what this load class can do. On 10 July 2024 a lightning arrester failure on a 230 kV line in the Eastern Interconnection produced six faults in 82 seconds, and about 1,500 MW of voltage-sensitive data center load transferred to backup power; frequency rose to 60.047 Hz [7]. On 22 July 2026 a conductor pulled free of its compression sleeve on a transmission line in the same interconnection, and about 3,869 MW of data center load transferred to backup power in two blocks sixty-seven seconds apart; frequency peaked at 60.092 Hz [12]. The Eastern Interconnection's resource loss protection criterion for 2026 is 3,875 MW. A single routine equipment failure removed almost exactly that much load, as a common-cause response across many facilities. Both were over-frequency events, and this paper does not claim that under-frequency load shedding operated in either. Their relevance here is that they show large blocks of load responding together to an ordinary fault at a scale comparable to the largest generation contingency the interconnection is designed for, in an environment where lower inertia makes any excursion steeper.

The regulatory response has followed. NERC's Level 3 Essential Action Alert of 4 May 2026 on computational loads covers modeling, studies, instrumentation, commissioning, operations,

protection and control, with responses due on 3 August 2026 [7]. On 16 July 2026 FERC directed NERC to file, by 31 December 2026, reliability standards on computational load integration together with registration criteria for those loads [13]. These are the regulatory anchoring events for the standards pathway developed in Section 7: the problem class is now formally identified on both sides of the balance, and the question before the standards community is what protection coordination framework is appropriate in response.

The combination of these two trends — lower and more variable synchronous inertia from IBR penetration, and more frequent large-block frequency perturbations, in both directions, from hyperscale load dynamics — creates a threat environment for which fixed PRC-024 settings are increasingly poorly calibrated. The misalignment between static protection settings and a dynamically variable inertia environment is the structural gap that this paper's proposed provisions pathway addresses. It also, incidentally, creates a more data-rich environment for the GP calibration architecture: more frequent disturbance events mean more FWHM observations, faster model convergence, and a training dataset that reflects the current inertia regime rather than a historical average.

3. Physical Foundation and Grid Application

3.1 Established Physical Basis

The principle that the Full Width at Half Maximum of a propagating wavefront encodes properties of the medium through which it has traveled is established across multiple branches of physics, and in each case rests on the same underlying mechanism: dispersion or attenuation accumulates along the propagation path. The temporal or spectral width of the observed feature at the downstream measurement point reflects that accumulated effect.

In spectroscopy, thermal Doppler broadening produces a Gaussian absorption line profile whose FWHM is proportional to the square root of the radiating medium's temperature. Pressure broadening produces a Lorentzian profile whose FWHM encodes the collision frequency of the absorbing species and hence the local gas density. The convolution of both mechanisms — the Voigt profile — carries information about both temperature and pressure simultaneously [1]. In quantitative spectroscopy, the FWHM of the observed line profile is not merely a descriptor of line shape; it is the primary observable from which the medium's physical state is extracted.

In a dispersive single-mode optical fiber, a short pulse broadens temporally as it propagates, and the temporal FWHM of the output pulse encodes the fiber's group velocity dispersion parameter integrated over the path length [2]. In ultrasonics, pulse FWHM broadening in viscoelastic media encodes frequency-dependent attenuation coefficients — and hence the medium's viscosity and elastic moduli — and underlies tissue characterization in diagnostic ultrasound and material characterization in industrial non-destructive evaluation [3]. In each domain, FWHM is a path-integrated quantity: it accumulates information about the medium along the entire propagation corridor, not solely at the observation point.

The formal analogy between these established results and frequency wavefront propagation in large power systems was drawn by Thorp, Seyler, and Phadke [4], who demonstrated that frequency disturbances propagate through the bulk electric system as electromechanical waves. Using a continuum approximation of the network, they showed that propagation characteristics are governed by the spatial distribution of machine inertia and network impedance — directly analogous to the group-velocity dispersion in a fiber or the viscoelastic attenuation in an acoustic medium. The inertia distribution of the grid serves as the medium's dispersive character; a downstream PMU's frequency observation carries information about the inertia environment integrated over the propagation corridor from the disturbance's origin. Co-author Cummings documented the wave-like propagation of the February 2008 South Florida frequency disturbance across the entire Eastern Interconnection using PMU data — the first interconnection-wide empirical observation of this phenomenon [11] — providing direct evidentiary grounding for the Thorp et al. framework in a real-world event.

The contribution of this paper is the application of this established physical principle to the protection coordination problem, with a specific and direct connection to PRC-024's ride-through requirement. FWHM is itself a duration — measured in seconds from the PMU frequency time-series — expressing the width of the frequency deviation event at half its maximum amplitude. The PRC-024 ride-through requirement is also expressed in duration: the time a unit must remain connected at a frequency below a specified level. The GP model therefore learns a duration-to-duration mapping, from the FWHM of the incoming wavefront to the predicted duration of the excursion below the ride-through threshold. Both quantities are measured in seconds, from the same PMU recording, without intermediate estimation steps. For a frequency excursion with approximately Gaussian temporal profile — the shape that propagation through an inertia-distributed medium tends to produce [4] — FWHM and threshold-exceedance duration are monotonically related by a transformation that depends only on the ratio of the threshold level to the maximum frequency deviation. The GP learns this relationship locally, without requiring the transformation to be derived analytically for a specific network topology. The domain-specific claim is that this mapping is sufficiently stable at a given PMU location under typical operating conditions that a locally calibrated GP model can reliably exploit it for ride-through decisions.

3.2 Physical Motivation

The physical basis for the FWHM-to-duration encoding follows directly from Thorp et al.'s electromechanical wave framework [4]. A synchronous machine with high inertia resists acceleration and deceleration — it exchanges kinetic energy with the grid slowly and acts as a damping element on the propagating wavefront. A propagation corridor containing machines with high aggregate inertia therefore slows and broadens the wavefront, producing a longer-duration frequency deviation at downstream observation points compared to what the same disturbance would produce propagating through a low-inertia corridor. Critically, a broader wavefront means the frequency excursion at the observation point is not only wider in FWHM

— it is also attenuating faster and recovering sooner. The unit experiencing a broad, attenuating wavefront is in a fundamentally different situation from one experiencing a sharp, sustained one: the former is likely to ride through; the latter may not. FWHM captures this distinction directly, in the same temporal units as the PRC-024 ride-through requirement.

The FWHM of the observed wavefront integrates this inertia effect over the entire propagation corridor. It is not a point measure of local conditions — it is a path-integrated proxy for how the medium has dispersed the disturbance between the origin and the PMU. This property makes it analytically distinct from ROCOF, which estimates inertia from the local machine's instantaneous swing dynamics and contains no information about the propagation corridor or the excursion's likely duration.

A secondary physical effect reinforces the expected relationship: load damping. High-load corridors, which tend to coincide with high-inertia transmission regions, also exhibit stronger load-frequency damping. Both mechanisms — machine inertia and load damping — broaden the wavefront and shorten the excursion duration in the same direction. The FWHM signal integrates both effects.

The growth of IBR penetration introduces a scoping condition that must be stated explicitly. Grid-following IBRs do not contribute to electromechanical wave propagation — they do not exchange kinetic energy through rotating mass coupling and therefore do not participate in the wavefront dynamics that produce the FWHM signal. FWHM, as used in this paper, denotes the synchronous inertia within the propagation corridor. In a high-IBR environment, synchronous inertia may be a conservative lower bound on total effective inertia — grid-forming IBRs contribute synthetic inertia that partially supplements the synchronous component — but the FWHM signal does not capture that synthetic contribution directly. For a protection application, this conservatism is in the correct direction: ride-through extension activates only when synchronous inertia alone is sufficient, without crediting synthetic inertia whose availability depends on IBR dispatch. We note that whether grid-forming IBR synthetic inertia leaves a detectable signature in the wavefront shape that a GP model could learn to identify is a follow-on research question and is deferred.

3.3 Why Closed-Form Derivation Is Not the Goal

Deriving an analytical FWHM-to-duration relationship for a general network topology requires solving the frequency wave equation on an impedance-distributed graph with spatially varying inertia and damping coefficients. This is not tractable in closed form for real network topologies: the wave equation is linear in a homogeneous medium, but the grid is neither spatially homogeneous nor topologically regular. Network impedances, machine characteristics, and load distribution introduce spatial heterogeneity, rendering the general solution analytically intractable.

The conjecture does not require solving this problem. It requires only that the encoding relationship exist and be learnable from local observations. This is the appropriate scope for the claim. The closed-form relationship is a path toward a universal model that generalizes across all network topologies. The conjecture and the learning architecture described in Section 4 are a path toward a locally calibrated model — one that knows this network, in this configuration, under typical operating conditions, and improves its estimates as it observes more local events. The locally calibrated model is more directly deployable and does not require solving the general problem.

3.4 The Local Encoding Hypothesis

The deployability hypothesis underlying the conjecture is that the FWHM-to-duration mapping, while not analytically derivable for general topologies, is sufficiently stable at a given network location — over a typical range of operating conditions — that a model observing accumulated local event data can learn a useful approximation. This hypothesis has three implications for the learning architecture.

First, the model is location-specific. A model trained on events observed at one PMU does not generalize without recalibration to a PMU at a different network location, because the inertia distribution relevant to a ride-through decision at any location is the inertia in the corridor between typical disturbance origins and that specific location.

Second, the model improves with experience. Every disturbance the PMU observes — with known outcome and known frequency trajectory — is a training sample that refines the model's estimate of the local FWHM-to-duration mapping. The model drifts toward local calibration as the event record accumulates.

Third, the model has a sensible prior. Before the model has observed any local events, the FWHM-to-duration relationship can be approximated from a simplified network model using available network parameters. This prior is not accurate — it ignores topology-specific effects — but it is physically motivated and, by design, conservative, providing a reasonable and safe starting point.

4. A Learning Architecture for Local Duration Estimation

4.1 Gaussian Process Regression

The learning architecture proposed for the local excursion duration estimation model is Gaussian process (GP) regression. A Gaussian process defines a distribution over functions — in this application, a distribution over possible FWHM-to-duration mappings. Given training data consisting of observed FWHM values and the associated measured excursion durations below the PRC-024 threshold, the GP updates this distribution to produce a posterior that concentrates on mappings consistent with the observed data.

GP regression was selected for three properties that are essential for deployment in a protection system context. First, explicit uncertainty quantification: at any input FWHM value, the GP posterior provides a full posterior distribution — mean and variance — from which the probability that excursion duration will fall within the ride-through window can be directly computed. This property is essential for the conservative activation logic described in Section 4.4. Second, interpretability: the GP model state at any time is fully inspectable; the kernel function and its learned hyperparameters are visible and auditable. Third, bounded behavior: the GP posterior is constrained by the prior and cannot drift arbitrarily from physically motivated bounds regardless of what the training data contains.

4.2 Kernel Selection and Physics-Based Prior

The GP kernel encodes the prior expectation about how similar FWHM values should produce similar predicted durations. A squared exponential kernel with a learned length scale is the natural choice: it asserts that nearby FWHM values produce similar duration estimates, with the length scale governing how rapidly the similarity decays. The kernel hyperparameters — amplitude and length scale — are initialized from the physics-based prior and updated as training data accumulates.

The physics-based prior encodes the expected FWHM-to-duration relationship before any local observations are made. It is constructed from the monotonic relationship described in Section 3.1: wider FWHM implies shorter excursion duration below threshold. The prior asserts this monotonically decreasing relationship, parameterized conservatively: for any observed FWHM, the prior mean assigns a duration estimate at the upper end of the physically plausible range for the local network class. This conservatism is intentional — a conservative prior means that before the model has accumulated sufficient local observations, the posterior duration estimate will exceed the ride-through time limit at the observed frequency, so the activation test of Section 4.4 will not be met and extension will not activate. As local observations accumulate, the posterior moves from the conservative prior toward the relationship actually observed at the location, and extension becomes available only where that evidence supports it.

This improvement property is bounded. The GP hyperparameters are constrained to remain within physically plausible ranges throughout the training process. Kernel amplitude is bounded by the plausible range of excursion durations for the network class; length scale is bounded by the range of FWHM values consistent with physically realistic wavefronts. The model cannot drift arbitrarily from the physics prior regardless of what the training data contains.

4.3 Training Data and Drift Toward Improvement

Each disturbance the PMU observes provides a training sample. The sample consists of: (1) the observed FWHM of the frequency wavefront, computed directly from the PMU frequency time-series; and (2) the measured duration of the frequency excursion below the applicable PRC-024 threshold at the local PMU, also read directly from the same PMU recording. No ROCOF

estimation, no post-event filtering, and no secondary calculation is required. The training label is a directly observable quantity in the same units as the input — both are durations in seconds from the same time-series. This simplicity is a significant architectural advantage: the ground-truth label no longer requires ROCOF-based estimation, which introduced filtering delay and measurement noise and was the weakest link in an inertia-mediated formulation.

As training samples accumulate, the GP posterior narrows. The model's uncertainty about the FWHM-to-duration mapping at the local PMU location decreases. The ride-through extension activates more frequently in high-inertia conditions because the posterior duration estimate falls within the ride-through window with greater confidence. The model drifts toward local calibration: it learns the relationship between wavefront width and excursion duration as it manifests at this specific network location, under the operating conditions typical for this facility.

4.4 Conservative Activation Logic

The following notation is used in this section: $FWHM_{obs}$ denotes the observed FWHM at the local PMU; μ_T and σ_T are the GP posterior mean and standard deviation of the predicted excursion duration at that FWHM value; k is a conservatism parameter set by the utility; and $T_{limit}(f)$ is the PRC-024 ride-through time limit at the observed frequency level f — read directly from the standard's frequency-time curve, not a learned parameter.

Extension activates when $\mu_T(FWHM_{obs}) + k \cdot \sigma_T(FWHM_{obs}) < T_{limit}(f)$. In plain terms: the model's predicted excursion duration, inflated by k standard deviations of its own uncertainty, must still fall within the PRC-024 ride-through window. Wide uncertainty inflates the upper bound, making it harder to remain below $T_{limit}(f)$, so extension does not activate. The architecture remains self-conservatizing: ignorance expresses itself as inaction.

Note that the conservatism threshold operates in the opposite algebraic direction from an inertia-based formulation. Estimating inertia requires subtracting uncertainty to ensure the lower bound exceeds a minimum threshold. Estimating excursion duration requires adding uncertainty to ensure the upper bound remains below a maximum limit. The underlying logic is identical — demonstrate safety under worst-case uncertainty — but the direction follows the output variable. The duration formulation has a further advantage: $T_{limit}(f)$ is not derived or estimated. It is read from the PRC-024 frequency-time curve at the current measured frequency, making the threshold unambiguous and fully auditable.

When k is large, the condition is difficult to satisfy — extension activates only when the model is highly confident that the excursion will resolve within the ride-through window. When k is small, the threshold is more readily met. The default value of k should be set conservatively during initial deployment and relaxed only as the model accumulates a validated event record. In all cases where the condition is not satisfied — including all cases where model uncertainty is wide — the existing PRC-024 protection setting governs.

4.5 Verification, Validation, and Auditing Requirements

Continuous model updating introduces a verification and validation challenge that must be addressed in the standards pathway. The proposed V&V framework has four elements.

Parameter bounds enforcement. GP hyperparameters are bounded within ranges derived from physically plausible network models for the local facility class. Updates that would push parameters outside these bounds are rejected. The bounds are specified at deployment and are not subject to field modification without a formal protection settings change process.

Holdout validation. A fraction of observed events — nominally twenty percent — is withheld from training and used to validate the model's duration predictions against directly measured excursion durations from the withheld events. The model's root mean square error on the holdout set is logged continuously. If holdout RMSE exceeds a utility-specified threshold, ride-through extension activation is suspended automatically pending manual review by the protection engineering staff.

Auditability. The model state — current kernel parameters, training data summary statistics, posterior mean function, holdout validation results, and the activation log — is continuously logged and available for inspection. The protection engineer can see exactly what the model has learned and review every instance in which extension was or was not activated.

Update rate limiting. Model parameter updates are performed on a scheduled basis — daily or weekly, depending on disturbance frequency at the location — rather than in real time following each event. Each update batch is subject to the holdout validation check before parameters are committed. Real-time update is explicitly excluded from the architecture.

4.6 Local Execution, and Where Local Sovereignty Sits

The architecture described in this section satisfies a design requirement that is distinct from, and more fundamental than, the V&V requirements in Section 4.5. A UFLS event is by definition a grid stress condition in which normal operational dependencies have been compromised. External power may be degraded. Wide-area communications may be congested or unavailable. Control room operators may be managing multiple simultaneous events. Any protection function whose decision depends on external power, external communications, or operator action is precisely the kind of function that may fail under UFLS conditions.

The adjustable UFLS PMU architecture described in this paper is designed for local execution: it must be capable of making and executing its ride-through decision without requiring external power, external communications, or operator action in the decision loop. Every element that touches the real-time protection decision is locally resident and locally powered: the FWHM computation runs on the local PMU's processor, the GP posterior is resident in local memory, the activation threshold comparison is a local arithmetic operation, and the ride-through

extension, if activated, is a local relay action. Nothing in the decision path requires a signal from outside the device.

Local execution is not local sovereignty, and an earlier version of this paper claimed more than the architecture can bear. Local sovereignty is defined in the Foundational Definition FD-LS [14]. Its physical-law condition requires that the protective decision be a fixed function of measured quantities and setpoints, deterministic within a stated input bound, formally testable as deployed, and free of any learned or stochastic element. The extension decides on a Gaussian process posterior learned from event data. That is inference, not measurement against a setpoint, and the extension fails the physical-law condition by design. It fails a second condition as well. Its parameters are updated on a schedule from observations that reach it over C37.118, and FD-LS requires that nothing on the reachable side can alter what a protective function acts on except through a governed act by a qualified human. Neither failure is a defect in the extension as an engineering proposal. Both mean that this paper may not describe the extension as locally sovereign.

What can carry the property is the protection the extension sits above. The extension is strictly additive and bounded: it activates only where the predicted excursion falls inside the PRC-024 ride-through window, and the time limit it tests against is read from the standard's frequency-time curve rather than learned. A protective element that trips on locally measured frequency at that fixed limit is a fixed function of a measurement and a setpoint, and it can meet all three conditions of FD-LS, provided its settings can be changed only through a governed act and nothing the model computes can withhold or delay its trip. The design requirement this paper therefore states is that the fixed-threshold floor be held in an element of that kind, which no activation of the extension can carry the unit past. That is a stronger architecture than a locally sovereign extension would have been, because the extension may fail, be corrupted or drift, and the floor does not depend on it. Whether any action of the extension falls above the Bright Line is decided on the consequence of that action, and the floor's local sovereignty does not answer it.

This is not a novel design principle in protective relay engineering. The Aurora protection device — developed to protect rotating machinery against the out-of-synchronism damage mode demonstrated in the 2007 Idaho National Laboratory test — was designed on the same principle. The attack scenario is precise: the adversary opens a breaker and intends to reclose it onto an out-of-synchronism grid in approximately 15 to 20 cycles, inducing the rotational stress that can destroy a large generator. The Aurora device reads the local grid physics and, when it detects an out-of-synchronism condition, lifts the close coil lead — making reclose mechanically impossible without any trip command, operator action, or communication with a remote system. The protection lies in preventing the injurious state transition, not in responding after it has occurred. The device works because it is purely local: it observes what the local grid is doing and acts on that observation alone.

The UFLS ride-through architecture shares the Aurora device's principle of local execution, in mirror-image logic. Where the Aurora device prevents an injurious close, the UFLS device

prevents an unnecessary open. Both read local physics — the Aurora device reads grid synchronism; the UFLS device reads wavefront shape — and both act locally to prevent a bad state transition from occurring. Local execution is the answer to threat conditions, whether adversarial or operational, in which the normal dependency chain cannot be relied upon.

UFLS conditions are the frequency stability analog. When frequency falls below UFLS thresholds, the grid is already under severe stress. The protection decision must be made in the first hundreds of milliseconds. Communications congestion, operator workload, and potential power supply degradation at the substation level are all elevated. A ride-through extension architecture that depends on any of these factors in the decision loop is not fit for purpose in the conditions it is designed to address. Local execution is not optional for this application — it is the enabling condition that makes the architecture credible.

The implications for standards are developed in Section 7.3. For the architecture, the consequence is a clear and non-negotiable constraint: the GP model, the activation logic, and the relay action must all be executable from local power and local state, with no real-time external dependency. The C37.118 calibration loop, the holdout validation, and the audit logging are all post-event processes that may use external communications — but none of them are in the decision path.

5. PMU-PMU Observation Sharing as Calibration Data Expansion

5.1 The Calibration Data Constraint

A significant practical constraint on the GP model is the rarity of large disturbances at any given network location. The model’s calibration quality depends on the number of training samples — FWHM observations with the associated measured excursion durations — accumulated at the local PMU. Events sufficient to produce a measurable FWHM signature may occur at a given location only a few times per year. At that rate, local-only model calibration converges slowly, and the period during which the model operates near its conservative prior — with wide uncertainty and infrequent activation of extensions — is prolonged.

5.2 C37.118 as Training Data Transport

IEEE C37.118 is the standard protocol for synchrophasor data communication in wide-area measurement systems. Every PMU in a utility’s measurement network transmits frequency, voltage, and current phasor data at 30 to 60 frames per second to a phasor data concentrator (PDC) via C37.118. This infrastructure is already deployed across the transmission system for monitoring and post-event analysis. No new communication infrastructure is required for the PMU-PMU sharing concept.

The proposal is to use this existing infrastructure as a training data transport for the GP calibration loop. When a disturbance occurs, every PMU in the measurement network that

observes a measurable frequency event records a FWHM value at a different point on the propagating wavefront. Those FWHM observations, paired with the excursion durations measured at each PMU, are training samples. Shared via C37.118 after the event, they expand the local model's training dataset with observations from disturbances that may have produced a stronger FWHM signature at other network locations than at the local PMU.

5.3 The Calibration Loop Is Not the Decision Loop

A critical architectural distinction separates the real-time protection decision from the calibration loop. The ride-through extension decision is made locally, in milliseconds, from the FWHM of the locally observed frequency wavefront and the current GP posterior. No C37.118 traffic is in the decision loop. No neighbor PMU data is required in real time. The latency characteristics of C37.118 transmission over utility-wide-area networks — typically 100 milliseconds or more — are irrelevant to the protection decision.

Neighbor PMU data arrives in the calibration loop after the event. Post-event FWHM observations from neighboring PMUs are transmitted to the local PDC, aggregated, and processed on the training data update schedule. They expand the local model's training dataset without introducing any dependency on communication latency in the decision path. The local decision remains fully local.

5.4 What Neighbor Data Adds to Model Quality

The value of neighbor PMU observations lies in the spatial coverage of the inertia landscape surrounding the local PMU. A disturbance originating near a distant neighbor PMU may produce a FWHM signal too small to be useful at the local PMU — the disturbance has attenuated over the long propagation path — while producing a clear signal at PMUs close to the origin. Those close-in PMU observations, shared via C37.118, inform the local model about how disturbances from that direction propagate through the intervening network.

Over many events from many directions, the local model accumulates a spatial picture of the effective inertia landscape. The improvement is not uniform: disturbances from directions well-represented in the training data calibrate the model more precisely than disturbances from novel or rare directions. The uncertainty quantification in the GP posterior reflects this: well-represented FWHM regimes have narrow posterior uncertainty, whereas poorly represented regimes retain wider uncertainty and are less likely to trigger the conservative activation threshold. The architecture self-corrects toward conservatism in situations where the model has less evidence.

6. The Flagami Case Study

6.1 Event Summary

On February 26, 2008, a fault at the FPL Flagami 345 kV substation in Miami, Florida, occurred while a field engineer diagnosing a malfunctioning switch had disabled two levels of protection on energized equipment, so the fault was not cleared locally. The resulting disturbance cascaded across the southern Florida peninsula. Both nuclear units at Turkey Point, about 40 miles south of Flagami, tripped, along with multiple fossil generating units. In all, 22 transmission lines, about 4,300 MW of generation and about 3,650 MW of customer load were lost, and nearly one million customers were interrupted, some for several hours [8].

The event was analyzed by NERC and the Florida Reliability Coordinating Council and investigated by FERC with NERC under Docket IN08-5, which concluded with civil penalty settlements, including $25 million from FPL in October 2009 [8]. The working premise of this case study is that inertia conditions in the affected region were consistent with successful ride-through for some of the generating units that tripped on frequency protection, meaning that those units need not have tripped to protect themselves. That premise is drawn from the co-authors' reading of the post-event record and must be confirmed against the analysis documents themselves before the simulation results are reported.

This premise is the evidentiary foundation for the Flagami case study as applied to the conjecture. The validation proposed here would not attempt to demonstrate that the conjecture's algorithm would have prevented the cascade. It would ask a more limited question: is the FWHM signature of a simulated frequency wavefront at the tripped unit locations, at the moment the protection decision would have been made, consistent with what a calibrated GP model would have recognized as a high-inertia condition?

6.2 Proposed Validation Method

A reconstruction of the Florida network would be built from NERC's power flow and dynamics model data, calibrated to 2008 system conditions using the load flow data and generator dispatch records in the post-event documentation, and run on any transient stability platform able to report frequency at sub-second resolution. The fault would be initiated at the Flagami 345 kV bus, consistent with the documented event, and the simulation would capture the resulting voltage excursion, the frequency transient across the Florida network, and the cascading unit trip sequence. The method is stated so that any group holding a suitable model can run it; this paper does not.

Frequency time-series would be extracted from the simulation at each generating unit location at PMU-equivalent sampling rates of 60 frames per second. FWHM would be computed from the extracted frequency time-series at each location using the following procedure: the frequency deviation from the 60 Hz nominal is computed; the maximum deviation is identified; the half-maximum level is established; and the duration of the frequency event at the half-maximum level — the interval between the two time points at which the deviation crosses the half-maximum level — is measured. That FWHM value would be the input to an illustrative GP model inference.

Any such reconstruction should be reported as an approximation, not as a validated model. The network model would approximate 2008 conditions, FWHM would be computed from simulated rather than measured frequency time series, and the GP model would be illustrative rather than trained on actual Florida grid-event data. The purpose is to test consistency with the conjecture, not to establish the GP model's readiness for deployment.

6.3 Consistency Assessment Framework

The consistency assessment would ask whether the simulated FWHM values at the generating unit locations fall within the range predicted by the conjecture for high-inertia conditions at the time of the event. Specifically, it would compare the simulated FWHM values at units that post-event analysis identified as having inertia conditions consistent with ride-through with those at units that clearly required tripping. If the conjecture is correct, the former should produce systematically larger FWHM values than the latter.

The GP model used for the assessment would be initialized from the physics-based prior with no local training data, representing the model's state at initial deployment. The posterior duration estimates at the observed FWHM values would be compared against the conservative activation threshold. The consistency question is: does the posterior estimate, even under wide prior uncertainty, indicate an excursion short enough to ride through at the units identified as ride-through-capable in the post-event analysis?

6.4 Status

No simulation has been run, and none is in prospect from the authors. The paper therefore reports no quantitative Flagami results and makes no claim that the FWHM signature at the tripped units supports the conjecture. The method above is offered for any utility, laboratory or vendor with a suitable dynamic model, and field PMU records from events with well-documented outcomes would serve the same purpose with measured rather than simulated wavefronts. Until one or the other is done, the conjecture rests on the physical basis in Section 3 and the architecture in Section 4, and the standards pathway in Section 7 should be read as contingent on that validation.

7. Standards Pathway

7.1 PRC-024 Current Structure

NERC Reliability Standard PRC-024 establishes frequency and voltage protection setting requirements that let generating resources ride through excursions. The current version, PRC-024-4, was approved by FERC in Order No. 909, effective 28 August 2025, and now applies to synchronous generators, synchronous condensers and Type 1 and Type 2 wind resources. Ride-through for inverter-based resources moved at the same time to a separate performance-based standard, PRC-029-1 [6]. PRC-024-4 requires that protective relays not trip a

unit while frequency remains within the standard's frequency-time curves, with separate curves for underfrequency and overfrequency conditions, drawn from the historical excursion profile of the North American interconnections. The architecture proposed here reads synchronous inertia through the wavefront, so its natural home is PRC-024 rather than PRC-029.

The standard's compliance architecture is fixed-threshold: a facility demonstrates compliance by demonstrating that its protection settings do not trip within the ride-through envelope. There is no provision for a site-specific, condition-adaptive, or model-based approach. The envelope is conservatively calibrated — for the worst-case inertia conditions the interconnection might present — and is applied uniformly regardless of the actual inertia conditions at the time of any specific event.

7.2 The Gap the Provisions Pathway Addresses

Fixed-threshold compliance does not distinguish between low-inertia and high-inertia conditions. The PRC-024 envelope was calibrated for conditions where generating unit trips would worsen the frequency excursion — conditions of low aggregate inertia, where each additional trip deepens the frequency decline. When aggregate inertia is high, the frequency excursion is more likely to be self-correcting, and trips of generating units within the PRC-024 envelope represent a lost ride-through opportunity without providing a protective benefit.

The result is a systematic conservatism bias that is most costly precisely when inertia is highest: under conditions where the grid is most resilient, protection settings engineered for worst-case inertia still trip units unnecessarily. As renewable generation displaces synchronous machines, aggregate inertia becomes more variable by hour, season, and dispatch condition. The variance in inertia conditions is increasing; the fixed PRC-024 envelope was calibrated for a lower-variance environment.

The 2026 federal actions on computational loads provide current institutional grounding for the gap argument. NERC's Level 3 alert of 4 May 2026 and FERC's order of 16 July 2026 both rest on the recognition that the existing framework was written for a generation-dominated disturbance profile and must now be examined in light of large loads and a changing inertia environment [7], [13]. A Standards Authorization Request filed while NERC is developing the computational load standards FERC has directed has a stronger institutional rationale than one filed without that anchor. It should cite both actions, and it should say how model-based ride-through for generators relates to the ride-through obligations now being written for loads, since both address the same excursions from opposite sides of the balance.

7.3 Proposed Provisions Pathway Structure

The provisions pathway within PRC-024 would allow a generating facility to demonstrate compliance through a model-based approach—specifically, a calibrated GP model that meets specified V&V requirements—as an alternative to fixed threshold settings. Under this approach,

the facility's protection system is never less protective than the existing PRC-024 envelope: the model-based extension is strictly additive. In high-inertia conditions where the GP posterior satisfies the conservative activation threshold, the facility remains connected longer, improving system reliability. In all other conditions, the existing envelope governs.

The local execution architecture described in Section 4.6 is a prerequisite for the provisions pathway, not merely a desirable feature. A model-based compliance demonstration that depends on external communications or operator action in the real-time decision loop cannot be relied upon under grid-stress conditions in which PRC-024 ride-through requirements apply. The provisions pathway must therefore include two mandatory architectural conditions. The first is local execution: the ride-through extension decision and its execution must be achievable from local power and the local state alone. The second is a locally sovereign floor: the fixed-threshold protection beneath the extension must meet the conditions of FD-LS [14], so that no activation of the extension, and no fault in it, can carry the unit past the PRC-024 limit. The extension itself is not locally sovereign, and the pathway should not describe it so. These requirements distinguish the proposed architecture from supervisory override schemes and settings management systems, both of which introduce external dependencies in the decision path.

Four conditions are required to claim model-based compliance under the provisions pathway:

Model qualification. The GP model meets specified accuracy requirements — holdout RMSE below a threshold calibrated to the facility's consequence severity — as documented in an annual validation report submitted to the regional reliability coordinator.

Conservative activation threshold. The model's activation logic is set such that extension never activates unless the posterior predicted excursion duration, inflated by the conservatism factor k, remains below the PRC-024 ride-through time limit at the observed frequency, as specified in Section 4.4.

V&V requirements. The facility maintains the parameter bounds enforcement, holdout validation, auditability, and update rate limiting described in Section 4.5.

Non-degradation. The model-based approach produces protection behavior no less conservative than the existing PRC-024 fixed envelope in expectation across the historical disturbance record for the facility's network location, as demonstrated by a retrospective analysis over the training event record.

7.4 Standards Development Pathway

The near-term deliverable is a NERC Standards Authorization Request (SAR) proposing revision of PRC-024-4 to include a provisions pathway for inertia-conditioned ride-through extension. The SAR would be filed with NERC's Standards Department and would initiate the standards development process under NERC's reliability standards development procedure. This paper supplies the physical and architectural basis for the SAR. A SAR would be stronger with the

validation proposed in Section 6, or with a field demonstration, and the evidentiary threshold NERC would require is one of the questions put to reviewers.

The appropriate drafting team for PRC-024 revisions is the Protection System Reliability Standards Drafting Team. The SAR would request that the provisions pathway be developed as an alternative compliance path within a revised PRC-024 or as a companion standard.

The IEEE Power System Relaying and Control Committee (PSRC) is the appropriate standards body for relay-level implementation requirements — specifically, the V&V framework, parameter bound enforcement, and audit logging described in Section 4.5. Coordination between the NERC standards process and IEEE PSRC is advisable to ensure that the technical requirements at the relay level are consistent with the reliability requirements at the standards level. A joint technical paper submitted to the IEEE PSRC, concurrent with the NERC SAR, would initiate that coordination.

Conclusion

This paper advances two contributions to the frequency protection coordination problem, rooted in a conversation near the Flagami substation about the grid ringing like a bell.

The first contribution is the application of an established physical principle — that FWHM of a propagating wavefront encodes properties of the medium through which it has traveled — to the specific and direct problem of predicting excursion duration at a local PMU. FWHM is itself a duration. The PRC-024 ride-through requirement is expressed in duration. The GP model learns a duration-to-duration mapping from the width of the incoming wavefront to the predicted time at which the local frequency will remain below the ride-through threshold. Both quantities are measured in seconds, from the same PMU recording, without intermediate estimation. The Thorp et al. framework establishes the physical basis [4]; Cummings's PMU documentation of the South Florida disturbance provides the empirical grounding [11].

The second contribution is a Gaussian process regression framework that exploits that mapping. The GP model starts from a conservative physics-based prior, learns the local FWHM-to-duration relationship from accumulated event data, produces explicit uncertainty estimates, and activates ride-through extension only when the predicted excursion duration — inflated by a conservatism factor — remains within the PRC-024 ride-through window. PMU-to-PMU observation sharing via the existing C37.118 layer expands the calibration dataset without placing communication latency in the real-time decision path. The architecture executes from local power and the local state alone, a condition that must hold when UFLS thresholds are reached, and it sits above a fixed-threshold floor that can be locally sovereign where the extension cannot.

The Flagami event motivates the work: a well-documented cascade in which generating units tripped on frequency protection, some of them, on the paper's working premise, under conditions consistent with successful ride-through. Section 6 proposes how a reconstruction of that event

could test whether the FWHM signature at those units is consistent with the duration predictions the architecture would have produced. That test has not been run.

The translational objective is a provision pathway within NERC PRC-024 for model-based alternative compliance. The paper does not claim the GP architecture is ready for deployment. It claims that the physical foundation is established, that the duration-to-duration formulation is more direct and more tractable than an inertia-mediated one, that the V&V framework is adequate for protecting the credibility of protection engineering, and that the standards pathway is the right vehicle for the broader validation the concept requires. Two items carry the work forward: a validation of the conjecture against a reconstructed or recorded event, by simulation or field PMU data; and the NERC Standards Authorization Request, which opens the standards development process.

References

[1] Corney, A., *Atomic and Laser Spectroscopy*, Oxford University Press, Oxford, 1977. (Spectral line broadening — Doppler, pressure, and Voigt profiles.)

[2] Agrawal, G.P., *Nonlinear Fiber Optics*, Academic Press, San Diego, 1989. (Temporal pulse broadening in dispersive single-mode fiber; group velocity dispersion.)

[3] Sachse, W. and Pao, Y.-H., "On the determination of phase and group velocities of dispersive waves in solids," *Journal of Applied Physics*, vol. 49, no. 8, pp. 4320–4327, Aug. 1978. (FWHM broadening in viscoelastic media; attenuation coefficient encoding.)

[4] Thorp, W.A., Seyler, C.E., and Phadke, A.G., "Electromechanical wave propagation in large electric power systems," *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, vol. 45, no. 6, pp. 614–622, Jun. 1998. (Formal electromechanical wave propagation framework; inertia distribution as the dispersive medium.)

[5] Rasmussen, C.E. and Williams, C.K.I., *Gaussian Processes for Machine Learning*, MIT Press, Cambridge, MA, 2006. (Gaussian process regression; kernel methods; uncertainty quantification.)

[6] NERC, "Reliability Standard PRC-024-4: Frequency and Voltage Protection Settings for Synchronous Generators, Type 1 and Type 2 Wind Resources, and Synchronous Condensers," and "Reliability Standard PRC-029-1: Frequency and Voltage Ride-through Requirements for Inverter-Based Resources," approved by FERC in Order No. 909, 90 Fed. Reg. (29 July 2025), effective 28 August 2025.

[7] NERC, Incident Review: Considering Simultaneous Voltage-Sensitive Load Reductions, 8 January 2025; and NERC, Level 3 Essential Action Alert, Computational Load Modeling, Studies, Instrumentation, Commissioning, Operations, Protection, and Control, 4 May 2026.

[8] FERC, Florida Blackout, Docket No. IN08-5-000, including the Order Approving Stipulation and Consent Agreement with Florida Power & Light Company, 129 FERC ¶ 61,016, 8 October 2009; and the NERC and Florida Reliability Coordinating Council analysis of the February 26, 2008 disturbance.

[9] IEEE, "IEEE Standard for Synchrophasor Measurements for Power Systems," IEEE Std C37.118.1-2011, IEEE Power and Energy Society, New York, 2011.

[10] Kundur, P., *Power System Stability and Control*, McGraw-Hill, New York, 1994. (Swing equation; ROCOF-based inertia estimation; frequency stability fundamentals.)

[11] Cummings, R.W., "System Disturbance and Blackout Analysis," *IEEE Power & Energy Magazine*, vol. 21, no. 3, pp. 30–43, May/Jun. 2023. DOI: 10.1109/MPE.2023.3247050. (First interconnection-wide PMU observation of electromechanical wave-like propagation; South Florida disturbance frequency wavefront documentation.)

[12] PJM Interconnection, Operating Committee, 6 August 2026, agenda item 5, presentation on the 22 July 2026 load transfer event; see also NERC, 2025 Frequency Response Annual Analysis, for the Eastern Interconnection resource loss protection criterion.

[13] Federal Energy Regulatory Commission, Order Directing NERC to File Reliability Standards Pertaining to Computational Load Integration, Docket No. RD26-7-000, 16 July 2026.

[14] Eclectic Technologies, “FD-LS: Local Sovereignty,” Foundational Definitions, October 2026 Edition, DOI 10.5281/zenodo.23117612.

Revision History

Version	Date	Description
0.8	3 October 2026	Conformed to the October 2026 Edition of the Foundational Definitions, whose FD-LS states the physical-law condition as a fixed function of measured quantities and setpoints with no learned or stochastic element. Section 4.6 is retitled and no longer says the extension meets the three conditions of local sovereignty: it separates the local execution the architecture achieves from local sovereignty, finds that the extension fails the physical-law condition by design and the reachable-update limb through its scheduled C37.118 calibration, and states as a design requirement that the fixed-threshold PRC-024 floor be held in a locally sovereign element the extension cannot carry the unit past. Section 7.3 makes local execution and a locally sovereign floor two separate mandatory conditions, and the Conclusion follows. Reference 14 names the edition. An erratum before the Introduction names the version 0.7 deposit and quotes the withdrawn sentence. The cover carries the concept DOI until the version DOI for this version is reserved. Publication handling under the CR-PB draft: version 0.7 is DOI-bearing and stays frozen; this version is the successor and goes forward as a new Zenodo version under the concept DOI. No change to the physical argument, the learning architecture or the standards pathway otherwise. Project 5's reading of the extension under FD-LS is the source of the correction.
0.7	17 September 2026	Publication pass for the Zenodo deposit of September 2026, as an author-only preprint at the author's decision of 17 September. The cover carries the citation block with the reserved version DOI, no longer states that the paper is submitted to a magazine, and no longer restricts distribution. The introduction's account of the 2009 conversation is the author's own; R. W. Cummings's 2023 paper remains cited as prior work at reference 11. Section 7.4 no longer names a standards contact. The reviewer questions appendix, which named reviewers and invited co-authorship, is removed. No change to the argument.
0.6	17 September 2026	Decisions of 17 September 2026 on the alignment register. Section 4.6 cites the Foundational Definition FD-LS for local sovereignty and shows the architecture meets its three conditions (new reference 14). The PNNL simulation will not be available, so Section 6 is recast as a proposed validation method: Sections 6.2 and 6.3 describe what a reconstruction would do, Section 6.4 states that no simulation has been run and no quantitative results are claimed, and the abstract, introduction, Section 7.4 and conclusion no longer promise results. The reviewer questions for PNNL are removed.
0.5	17 September 2026	Alignment run across the always-on paper family. No change to the argument or the citations, which name no companion paper by version. The revision table now lists the newest version first, under the author's 17 September 2026 direction.
0.4	17 September 2026	Alignment with the always-on paper family. Reference 7 completed as the NERC incident review of January 2025 and the Level 3 alert of 4 May 2026, and the unsupported statement that the alert documents UFLS activations removed. Reference 8 no longer cites the 2011 Arizona-

Version	Date	Description
		Southern California outage as a precedent for Flagami and now cites the FERC Florida blackout docket. Section 2.5 corrects the sign of a load-loss event, which raises frequency, and adds the July 2024 and July 2026 load transfer events and FERC's order of 16 July 2026 (new references 12 and 13). Section 6.1 corrects the Flagami figures to 4,300 MW of generation, 3,650 MW of load and nearly one million customers, and states the ride-through finding as a premise to confirm against the post-event record. Section 7 updates PRC-024-3 to PRC-024-4 and notes PRC-029-1. Sections 3.3, 3.4, 5, 6.3 and 7.3 now use the duration formulation consistently, and the truncated sentence in Section 4.2 is completed. Open before submission: the Flagami ride-through premise; the title and issuing body of the NERC and FRCC analysis cited in reference 8; and the July 2026 figures, which come from the regional transmission organization's presentation pending any NERC review.
0.3	May 2026	Draft for reviewer circulation.